

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MODERATE-priority dimension with concrete structural gaps. The 23-slot ontology was rigorously developed with merge consolidation [\[Q54, Q55, Q119\]](#) and includes deployment-relevant types (BANK_NAME, PAYMENT_COMPANY, BILL_TYPE, MONEY, ACCOUNT_TYPE, CITY_OR_PROVINCE, PLACE_NAME, DATE, TIME). However, the dimension fails on critical deployment requirements: dataset inspection confirms (D19) that no METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE, WOREDA, or sub-city slot types exist in any of 19 Amharic examples, and these are core to EEU prepaid top-up and AAWSA billing workflows. Nine 'infrequent slots' were eliminated [\[Q52\]](#) without documented utility-deployment justification. Cross-lingual annotation choices show ontological ambiguity: SNEL (DRC electricity utility) is annotated as PAYMENT_COMPANY (D28), conflating utility provider with payment channel — relevant if EEU/AAWSA appear similarly. The deployment elicitation flags this as MODERATE priority because BANK_NAME/PAYMENT_COMPANY/BILL_TYPE provide partial coverage, but the missing types are non-negotiable for the core workflow.

ID	Evidence
Q52	"Consequently, nine infrequent slots from NATIONALITY through PLUG TYPE were eliminated." (p.4)
Q54	"Appendix A.2 contains all 34 slot types selected." (p.4)
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)
D19	በወር ለውጥ ስንት ነው መክፈል የሚገባኝ? — No METER_ID, UTILITY_ACCOUNT_NUMBER, or STS_TOKEN slots present anywhere in Amharic sample
D28	"Niongo ya snel ezali 2000 FC Yebisa ngai soki nakosi" — PAYMENT_COMPANY: snel, MONEY: 2000 FC — Lingala: "SNEL bill is 2000 FC, tell me if I've paid" — SNEL is the DRC national electricity utility; utility company annotated as PAYMENT_COMPANY

Strengths

- 23 slot types include deployment-relevant categories: BANK_NAME, PAYMENT_COMPANY, BILL_TYPE, MONEY, ACCOUNT_TYPE, CITY_OR_PROVINCE, PLACE_NAME, DATE, TIME, NUMBER [\[Q144\]](#)
- Telebirr fits PAYMENT_COMPANY (D4) and BILL_TYPE accommodates electricity (D1) — partial coverage of deployment slot space
- Annotation guide provides explicit disambiguation rules (e.g., temporal: <1 day → TIME; ≥1 day → DATE [\[Q153, Q154\]](#); bank name excludes 'bank' unless attached [\[Q152\]](#))
- Slot consolidation 34→23 was driven by frequency analysis [\[Q51, Q119\]](#), reducing label noise

ID	Evidence
Q51	"We decided to exclude slot entities appearing less than 500 times across all languages (after MUSIC_GENRE in the figure)." (p.4)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)

Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q152	"When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
D1	"በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic
D4	"በቴሌብር የብድር አካውንቱ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Deployment-relevant present categories: BANK_NAME, PAYMENT_COMPANY, BILL_TYPE, MONEY, ACCOUNT_TYPE, CITY_OR_PROVINCE, PLACE_NAME, DATE, TIME, NUMBER [Q144].

OO-2: Confirmed missing for Ethiopian utility deployment (D19): METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN (20-digit prepaid token), KEBELE, WOREDA, sub-city, NEIGHBORHOOD_NAME (informal Addis Ababa toponyms). The annotation guide requires explicit named entities for PLACE_NAME [Q156], so informal kebele/neighborhood references may not even fit existing types cleanly.

OO-3: Some category encodings reflect non-Ethiopian assumptions: PAYMENT_COMPANY conflated with utility provider (D28: SNEL); SHOPPING_ITEM, MUSIC_GENRE, ARTIST_NAME, DISH_OR_FOOD, MEAL_PERIOD relevant only to non-utility domains. Nine slots (NATIONALITY through PLUG TYPE) were eliminated [Q52] — without justification of trade-off against utility-specific slots.

OO-4: Stakeholder-driven taxonomy redesign is required. EEU customer service agents and AAWSA billing staff should define slot schemas for METER_ID format, account-number format, kebele granularity, and STS token recognition.

OO-5: Structural validity issue: deployment-critical decision boundaries (meter ID extraction, STS token recognition, kebele resolution) cannot be expressed in the 23-slot taxonomy, so benchmark performance does not reflect deployment task structure. Content validity issue: 9 missing slot types are deployment-critical.

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Q156	"With this entity, only annotate if entity is named explicitly, e.g Name of airport, museum or mall is not and nt just "mall", "airport" etc" (p.24)

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Information Gaps

- EEU and AAWSA's actual slot schemas for meter IDs, account numbers, and addresses (not in public sources)
- Whether any pre-merge slot types (among the dropped 9) would have been useful for utility deployment

Requires Expert Verification

- EEU billing system's canonical slot taxonomy and how informal Amharic place references should be normalized
- AAWSA's account-number and address conventions

Remediation

Gap 1: 23-slot ontology lacks METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE, WOREDA, sub-city, and NEIGHBORHOOD_NAME types — confirmed absent in dataset (D19).

Recommendation: Co-design an extended slot schema with EEU and AAWSA billing staff; add at minimum METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, and KEBELE; re-annotate the existing Amharic split or annotate a deployment-specific augmentation under the extended schema.

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